

SEMIYARKA, Kazakhstan

WMO: 361520

Lat: 50.8927N

Lon: 78.3222E

Elev: 149

StdP: 99.55

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-32.5	-39.4	0.1	-36.0	-35.8	0.1	-32.3	12.0	-8.5	10.3	-7.6	1.8	140	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	34.1	18.2	32.2	17.8	30.4	17.3	20.0	28.6	19.3	28.0	18.6	27.2	3.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.9	22.0	16.6	12.0	21.7	15.5	11.2	21.6	58.1	28.1	55.7	27.9	53.4	27.3	34.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.3	9.7	8.3	DB	-36.9	37.5	5.0	1.7	-40.5	38.7	-43.4	39.7	-46.2	40.6	-49.8	41.8
(5)				WB	-37.0	21.8	4.9	1.5	-40.5	22.8	-43.4	23.7	-46.2	24.5	-49.8	25.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-16.9	-15.4	-5.2	7.6	14.5	20.9	22.1	20.2	13.5	5.8	-3.9	-12.3	(6)
(7)		DBStd	15.21	8.95	8.47	7.54	5.76	4.72	3.98	3.35	3.80	4.74	5.14	7.45	7.82	(7)
(8)		HDD10.0	3416	833	712	471	109	14	0	0	0	20	148	417	691	(8)
(9)		HDD18.3	5438	1092	945	729	323	136	21	7	24	157	389	667	949	(9)
(10)		CDD10.0	1352	0	0	0	38	153	328	376	316	124	18	0	0	(10)
(11)		CDD18.3	332	0	0	0	1	16	99	124	81	11	0	0	0	(11)
(12)		CDH23.3	3987	0	0	0	34	288	1154	1315	983	208	5	0	0	(12)
(13)		CDH26.7	1593	0	0	0	7	87	475	548	404	71	1	0	0	(13)
(14)	Wind	WSAvg	3.8	3.5	3.7	3.9	4.1	4.1	3.8	3.6	3.6	3.7	3.9	3.9	(14)	
(15)	Precipitation	PrecAvg	226	10	11	10	12	23	28	35	28	20	17	12	(15)	
(16)		PrecMax	378	36	27	43	41	76	75	86	91	48	36	32	(16)	
(17)		PrecMin	143	1	2	1	0	0	1	2	0	2	4	0	0	(17)
(18)		PrecStd	48	8	7	8	10	17	22	22	13	10	8	8	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	3.0	15.9	27.2	33.2	35.6	36.2	32.5	23.4	13.2	3.2	(19)	
(20)			MCWB	0.6	1.1	9.1	13.7	16.8	18.0	19.3	18.8	16.4	11.9	7.4	1.4	(20)
(21)		2%	DB	0.5	1.1	10.7	23.3	29.1	33.3	33.7	33.5	28.5	19.5	10.2	1.6	(21)
(22)			MCWB	-0.9	-0.5	5.6	12.4	15.4	17.5	18.7	18.2	14.7	10.4	6.3	0.3	(22)
(23)		5%	DB	-1.9	-0.8	6.5	20.3	26.7	31.3	31.8	30.9	25.4	16.6	7.1	0.1	(23)
(24)			MCWB	-3.2	-2.2	3.7	10.8	14.5	17.3	18.3	17.5	13.7	9.2	4.3	-1.3	(24)
(25)		10%	DB	-4.3	-2.9	3.8	17.7	24.3	29.4	29.8	28.7	22.5	13.7	4.8	-1.5	(25)
(26)			MCWB	-5.4	-4.0	1.8	9.7	13.7	17.0	17.8	16.8	12.4	7.8	2.6	-2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	1.5	9.3	15.1	17.9	20.6	21.4	20.4	17.8	13.1	9.3	1.9	(27)
(28)			MCDWB	2.1	2.7	15.3	24.2	27.4	29.2	30.2	31.3	29.4	20.4	11.6	3.0	(28)
(29)		2%	WB	-0.8	-0.5	6.1	13.1	16.6	19.4	20.2	19.3	15.7	11.2	6.4	0.4	(29)
(30)			MCDWB	0.5	0.9	9.7	21.7	27.0	28.8	28.0	29.4	25.6	18.3	9.7	1.6	(30)
(31)		5%	WB	-3.2	-2.1	3.7	11.6	15.4	18.6	19.6	18.5	14.4	9.6	4.4	-1.2	(31)
(32)			MCDWB	-1.9	-0.9	6.2	19.0	24.2	27.8	27.5	28.0	23.6	15.5	7.1	-0.1	(32)
(33)		10%	WB	-5.4	-4.0	1.9	10.1	14.3	17.8	19.0	17.7	13.3	8.2	2.6	-2.7	(33)
(34)			MCDWB	-4.3	-2.9	3.7	16.7	22.5	26.4	27.0	26.7	21.4	13.4	4.6	-1.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.7	9.9	12.9	13.6	12.6	11.8	13.0	12.9	10.4	8.0	9.2	(35)
(36)			MCDBR	9.4	10.0	11.6	17.6	17.6	16.1	15.2	16.8	18.0	15.6	10.4	8.6	(36)
(37)			MCWBR	8.6	8.9	7.6	8.6	7.2	5.3	4.6	5.6	7.6	8.1	7.3	8.0	(37)
(38)		5% WB	MCDBR	9.6	9.8	11.2	16.1	15.1	13.2	12.2	14.2	15.7	13.9	10.0	9.0	(38)
(39)			MCWBR	8.9	8.9	7.7	8.4	7.2	5.1	4.6	5.4	7.3	7.8	7.3	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.343	0.399	0.423	0.428	0.427	0.452	0.423	0.377	0.357	0.344	0.315	(40)	
(41)		taud	2.054	1.965	1.977	2.099	2.137	2.222	2.161	2.216	2.289	2.247	2.131	2.044	(41)	
(42)		Ebn at Noon	675	745	776	813	828	831	801	805	804	739	622	585	(42)	
(43)		Edh at Noon	83	120	145	144	145	135	141	127	105	91	78	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.38	2.51	3.83	4.83	6.14	6.49	5.92	5.30	3.95	2.24	1.18	0.95	(44)	
(45)		RadStd	0.12	0.18	0.30	0.35	0.43	0.37	0.41	0.25	0.31	0.28	0.13	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page